

Crystal Facet Engineering on SrTiO₃ Enhances Photocatalytic Overall Water Splitting

Yang Zhang, Xuefeng Wu, Zhi-Hao Wang, Yu Peng, Yuanwei Liu, Shuang Yang, Chenghua Sun, Xiaoxiang Xu, Xie Zhang, Jun Kang, Su-Huai Wei, Peng Fei Liu,* Sheng Dai,* and Hua Gui Yang*



Cite This: *J. Am. Chem. Soc.* 2024, 146, 6618–6627



Read Online

ACCESS |



Metrics & More

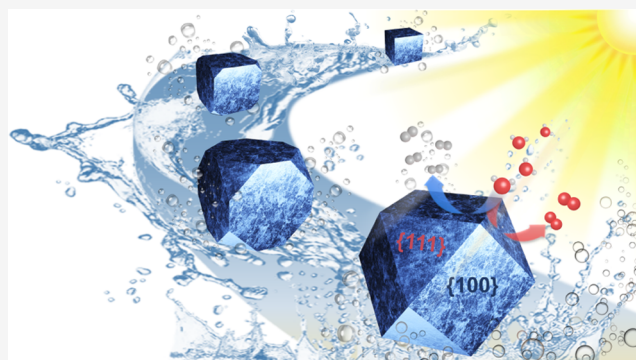


Article Recommendations



Supporting Information

ABSTRACT: Single-crystal semiconductor-based photocatalysts exposing unique crystallographic facets show promising applications in energy and environmental technologies; however, crystal facet engineering through solid-state synthesis for photocatalytic overall water splitting is still challenging. Herein, we develop a novel crystal facet engineering strategy through solid-state recrystallization to synthesize uniform SrTiO₃ single crystals exposing tailored {111} facets. The presynthesized low-crystalline SrTiO₃ precursors enable the formation of well-defined single crystals through kinetically improved crystal structure transformation during solid-state recrystallization process. By employing subtle Al³⁺ ions as surface morphology modulators, the crystal surface orientation can be precisely tuned to a controlled percentage of {111} facets. The photocatalytic overall water splitting activity increases with the exposure percentage of {111} facets. Owing to the outstanding crystallinity and favorable anisotropic surface structure, the SrTiO₃ single crystals with 36.6% of {111} facets lead to a 3-fold enhancement of photocatalytic hydrogen evolution rates up to 1.55 mmol·h⁻¹ in a stoichiometric ratio of 2:1 than thermodynamically stable SrTiO₃ enclosed with isotropic {100} facets.



INTRODUCTION

In view of increasingly severe global energy consumption, environmental issues, and climate change, the utilization of inexhaustible solar energy to generate renewable and clean hydrogen has become extremely important.¹ Among the technologies available for solar energy utilization, overall water splitting (OWS) using particulate photocatalysts provides a simple and cost-effective approach for green hydrogen production.^{1–3} Recently, a reaction panel array on a 100 m² scale using SrTiO₃:Al photocatalysts and a high solar to hydrogen (STH) efficiency up to 9.2% achieved by InGaN/GaN nanowire under a concentrated solar light demonstrated the substantial potential for large-scale solar to hydrogen conversion using one-step-excited crystalline photocatalysts.^{2,4} Over the past half-century, numerous light-harvesting materials (ranging from ultraviolet to near-infrared region) for OWS have been developed,^{5,6} including metal oxide,⁷ (oxy)nitride,⁸ (oxy)sulfide,⁹ g-C₃N₄,¹⁰ and hybrid perovskite.¹¹ Unfortunately, the photoexcited carriers involve complex charge separation, transfer, recombination, and surface (backward)-reactions over a wide time span from femtoseconds to seconds,¹² resulting in the low concentration of surface-reaching photocarriers and related slow redox reaction kinetics.¹³ Therefore, constructing high-quality photocatalysts

featuring efficient charge utilization and reactive surfaces is essential for high-efficiency solar hydrogen production.

Crystal facet engineering has been developed as an important strategy to tailor crystal surfaces with unique atomic configurations and coordination,¹⁴ which is crucial to promoting charge separation and transfer in the bulk photocatalyst as well as suppressing charge recombination at the surface and creating a suitable local environment for cocatalyst deposition and catalytic reactions. Various facet-dependent properties, such as photoreaction activity,¹⁵ electronic structure,¹⁶ built-in electric fields,^{17,18} redox reaction sites,^{19,20} interfacet junction,²¹ and photocorrosion resistance²² have been explored to boost photochemical reactions. In particular, a highly crystallized single-crystal SrTiO₃ enclosed with {100} and {110} facets prepared through SrCl₂ flux treatment exhibited a quantum efficiency of almost unity,⁷ giving a definitive prospect of crystallographic facet designing for OWS reaction. However, an advanced facet engineering

Received: October 29, 2023

Revised: January 30, 2024

Accepted: January 31, 2024

Published: February 13, 2024



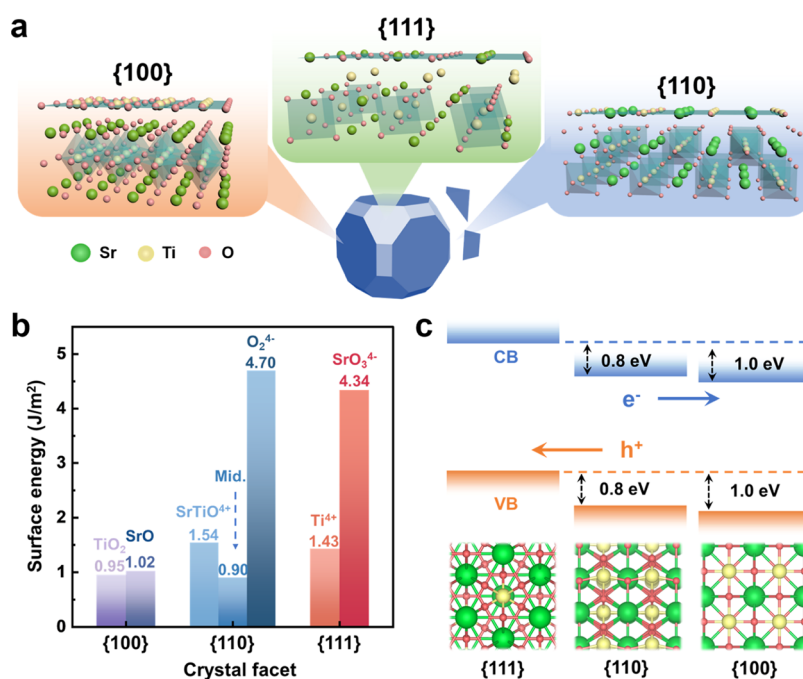


Figure 1. DFT studies of SrTiO₃ crystallographic facets. (a) A schematic diagram of SrTiO₃ single-crystalline model exposed with TiO₂-terminated {100} facet, SrTiO₃⁴⁺-terminated {110} facet, and SrO₃⁴⁻-terminated {111} facet, showing a unique atomic configuration with distinctive facets. (b) Surface energies of typical facets with various terminal atoms. (c) Calculated band-edge energy level of typical facets with mostly stable terminal atoms, suggesting the favorable charge separation between {111} and {100} facets. Noting: the termination of {110} facets shown in (c) is an intermediate case between SrTiO₃⁴⁺ and O₂⁴⁻. Sr, Ti, and O atoms are represented by green, yellow, and red spheres, respectively, in (c).

strategy to synthesize the well-defined single-crystal SrTiO₃ with tailored facets with regard to the OWS reaction, especially through solid-state synthesis with high crystallinity, is still unrealized and remains an important challenge.

Herein, we successfully develop a crystal facet engineering strategy through solid-state recrystallization to fabricate a highly crystallized SrTiO₃ single crystal exposed to tailored {111} facets. A unique presynthesized low-crystalline SrTiO₃ precursor was adopted to facilitate the formation of well-defined single crystals by improving solid-state structural transformation kinetics; the additive of Al³⁺ ions was demonstrated to tailor crystallographic facet structure, exposing {111} facet at a controllable percentage. Aberration-corrected scanning transmission electron microscopy and theoretical calculations identify the TiO₂-terminated {100} facets and SrO₃⁴⁻-terminated {111} facets with high surface energy, which can promote anisotropic charge separation and surface hole-trapping capacity. Notably, the photocatalytic activity for OWS is positively correlated with the exposure percentage of {111} facets, and the cuboctahedron SrTiO₃ single crystals with 36.6% of {111} facets exhibit hydrogen evolution rates up to 1.55 mmol·h⁻¹ in a stoichiometric ratio of 2:1, which is 3 times higher than isotropic cubic SrTiO₃ enclosed by {100} facets.

RESULTS AND DISCUSSION

Crystallographic Facet Studies via Density Functional Theory (DFT) Calculations. As a cubic crystal within the perovskite structure, the SrTiO₃ (STO) single crystal has three representative low-index surfaces, {100}, {110}, and {111}, and the respective schematic illustrations of the stoichiometric surface atomic configurations are shown in Figure 1a. The photoelectrochemical properties of STO single crystals are strongly dependent on the exposed surface and the

composition of the atomic termination layer.^{14,23} Figures 1b and S1 plot the relationship between the surface energy and the oriented surface with various terminations. Energetically, SrO- or TiO₂-terminated {100} surfaces with similarly low surface energies are the most favorable terminating facets, which are the typical surfaces of cubic STO. The lower surface energies for {110} and {111} facets are terminated with SrTiO₃⁴⁺ and Ti⁴⁺ atomic layers, respectively. However, the facets with a higher surface energy are generally more reactive than those with a lower surface energy.^{15,24} Although the {110} facet terminated with O₂⁴⁻ ({110}-O₂⁴⁻) has a significantly higher surface energy, it is unstable for the unsaturated coordination-rich layer of oxygen atoms as the outmost layer. Substantial reduction of surface energy from 4.7 to 0.9 J·m⁻² can be achieved by oxygen removal from the outermost layer of the O₂⁴⁻ layer, forming an intermediate state ({110}-mid), which might facilitate the stable existence of {110} crystal facets in practical situations. The {111} facet terminated with SrO₃⁴⁻ ({111}-SrO₃⁴⁻) also shows a much higher surface energy of 4.34 J·m⁻².

In addition, the different surface atomic configurations would induce changes in the Fermi energy level and thus thermodynamically guide the electron–hole pairs toward different crystalline surfaces.^{16,21} The position of the continuously changing energy band structure of typical facets with various terminations is plotted in Figure S2, using the {100}-SrO facet as a reference. As shown in Figure 1c, the energy of the {110}-mid facet falls in the energy position between the {111}-SrO₃⁴⁻ and {100}-TiO₂ facets, indicating that the photogenerated holes are more prone to be extracted to the {111}-SrO₃⁴⁻ facet enabling efficient oxidation reaction, and the {100}-TiO₂ facet serves as the reduction reaction-generating surface. Considering the high surface energy and excellent hole extraction ability, it is expected that optimizing

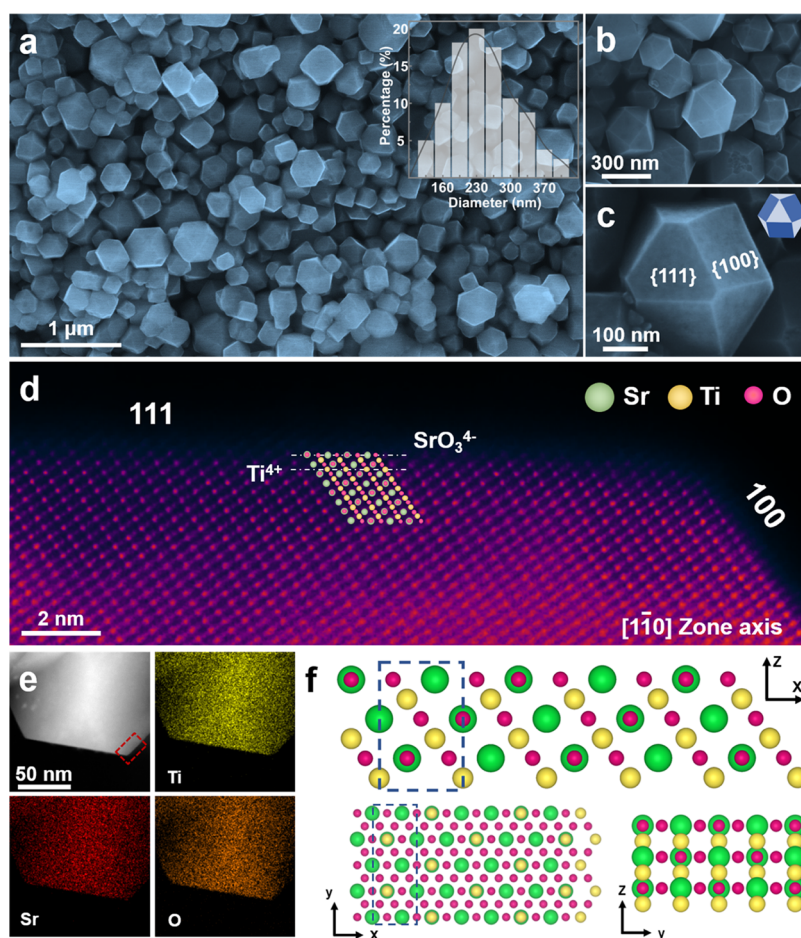


Figure 2. Crystal characterization of cuboctahedron SrTiO_3 single crystals. (a–c) SEM images of Cubo-STO, exhibiting widely regular single-crystal particles with well-defined and clean $\{100\}$ and $\{111\}$ facets. Inset: the corresponding particle diameter distribution, showing a vast majority of particle size in the 150–300 nm range. (d) Atomic-scale HAADF-STEM image along the $[1\bar{1}0]$ zone axis of Cubo-STO acquired from the red box in (e), showing an atomically flat surface. (e) HAADF-STEM image and the corresponding EDS elemental maps, suggesting the homogeneity of crystal components. (f) Atomic model of Cubo-STO in (c) projected from three vertical directions. Sr, Ti, and O atoms are represented by green, yellow, and red spheres, respectively, in (f).

the crystal structure would boost charge separation and reactivity by carefully modulating the orientation growth kinetics to selectively expose the $\{111\}$ and $\{100\}$ facets. Rohrer et al. have observed that the reduction and oxidation reactions occur on terraces with different terminations and opposite charges for the $\{111\}$ facet using mechanochemically polished STO.²⁵ So far, although STO single crystals enclosed by typical facets have been successfully synthesized via wet chemical synthesis,^{20,26} the $\{111\}$ facet is rarely exposed. Well-defined single crystals exposing a large fraction of $\{111\}$ surface have not been synthesized.

Morphology and Crystal Structure Investigation. A modified solid-state method using melt salt as reaction media was conducted to prepare the well-defined STO single crystals enclosed with $\{111\}$ and $\{100\}$ facets. A low-crystalline STO (Low-Cry-STO) nanosphere was subtly synthesized as a precursor.²⁷ The quenching method was adopted to obtain the low-crystalline nanospheres in a shorter reaction time in which the hydrothermal reaction autoclave at 180 °C was immersed directly into cold water for rapid cooling. To ensure the homogeneous dispersion of the Al^{3+} additive in molten salt, an impregnation method was adopted to mix $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ with STO nanosphere precursor ahead (more synthesis details are provided in the Supporting Information).

Scanning electron microscopy (SEM) images (Figures 2a–c and S3) show a cuboctahedron STO (Cubo-STO) crystal structure with regular morphology and smooth surface. Based on the schematic model as shown in the inset of Figure 2c, the eight triangular facets correspond to $\{111\}$ facets and the six square facets are $\{100\}$ facets of cubic phase STO. The percentage of the $\{111\}$ facet is determined to be 36.6% calculated from the geometric area. In addition, atomic-scale high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images are presented in Figures 2d and S4. The atomically flat surface configuration and the highly ordered atomic structure from surface to bulk were clearly observed on a large scale, indicating that the nanoparticle is a single crystal with good crystallinity. STEM-based energy-dispersive X-ray spectroscopy (EDS) elemental maps (Figures 2e and S5) illustrate the uniform distribution of Sr, Ti, and O with stoichiometric ratios. The insets in Figure 2d,f show the atomic model of stoichiometry STO in which the alternating SrO_3^{4-} and Ti^{4+} atomic layers are identified along the $[1\bar{1}0]$ zone axis. The HAADF-STEM image in Figure 2d clearly shows the $\{111\}$ surface terminated by the SrO_3^{4-} layer and the $\{100\}$ surface terminated by TiO_2 layers. It is expected that the dipole field generated by surface domains with uniform polarity and nonpolar terminations may induce an

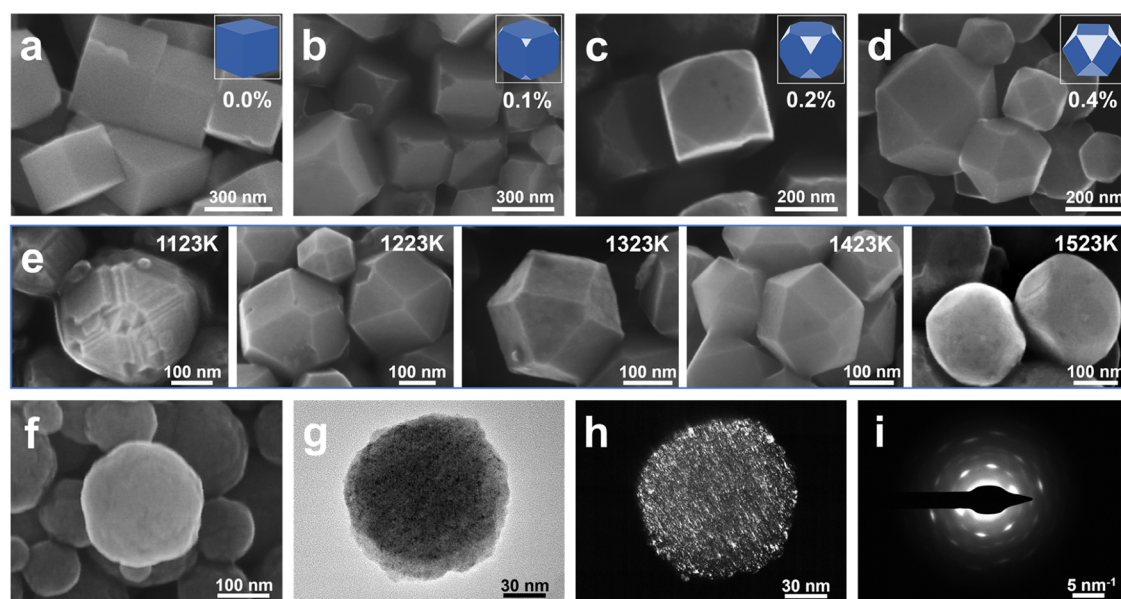


Figure 3. Facet engineering and well-defined structure modulating via solid-state synthesis. (a–d), SEM images of STO single-crystal particles with serial amount of Al doping (0.0–0.4 mol %) calcined at 1423 K from Low-Cry-STO, showing Al-dependent evolution of {111} facets. (e) SEM images of single-crystal STO calcined at 1123–1523 K, the well-defined cuboctahedron morphology was observed in the temperature range of 1223–1423 K. (f) SEM image, (g) bright-field TEM image, and (h) dark-field TEM image collected from the diffraction spots corresponding to {110} facet, and (i) corresponding selected area electron diffraction (SAED) patterns of Low-Cry-STO with a reaction time of 2 h, indicating the partial crystallization nanosphere.

internal electric field driving the anisotropic separation of photogenerated electrons and holes between {100} and {111} facets.

Aberration-corrected integrated differential phase contrast-STEM (iDPC-STEM) imaging was carried out to further identify the real position of O atoms which are not observed in the HAAF-STEM images.²⁸ The false-colored iDPC-STEM images of a typical Cubo-STO crystal are shown in Figure S6. Along the [110] zone axis, the interior Ti–O–Ti array consisted of Ti atomic columns and O atomic columns and the Sr/O column array composed of alternating O and Sr atoms were presented as a strict periodic arrangement from surface to bulk, consistent with the SrTiO₃ crystal structure. No visible structure distortion or disordered O positions were observed. Notably, the irregular terminating atomic configurations coexisting with SrTiO₄⁺ and an intermediate state (O₂^{4–} with partial oxygen loss) along the [110] orientation observed in Figure S6a are well consistent with our DFT analysis of surface energy.

Crystal Facet Engineering of Well-Defined SrTiO₃ Single Crystal. In the absence of kinetic control, the single-crystal structure synthesized by the high-temperature solid-state reaction can only be shaped by a thermodynamically stable surface.²⁹ With a high surface energy, the {111} facet terminated with SrO₃^{4–} should grow fast and finally vanish during growth. As a result, the polar {111} facet is energetically unfavorable to be exposed. Our group has previously demonstrated that adsorbate atoms can change the relative stability among surfaces.^{15,30,31} Herein, precision-controlled Al³⁺ ions were found to act as morphologically controlling agents. The exposure percentage of the {111} facet was strongly dependent on the amount of Al³⁺ additive (x mol % represents the Al/Ti molar ratio in the starting mixture). A series of single-crystal particles with different exposure percentages of {111} facets were investigated by SEM as

shown in Figures 3a–d and S7. Without an Al³⁺ additive, only cubic single crystals can be prepared through isotropic growth along the [100] orientation (Figure 3a). Interestingly, when 0.1 mol % Al³⁺ additive was added, pony-size {111} facets were exposed at the truncated corners (Figure 3b). The exposure percentage of the {111} facet develops with increasing amounts of Al³⁺ additive (Figure 3c), and the ultimate cuboctahedron structure (Figure 3d) forms with 0.4 mol % Al³⁺ additive, which would not further develop with further increasing Al³⁺ additive (Figure S7). Thus, the additive of Al³⁺ ions effectively modulated the surface structure of the crystal, selectively tailoring the {111} facet up to the fraction of 36.6%. The cuboctahedron structure was formed ranging from 1223 to 1423 K using Low-Cry-STO (Figures 3e and S8), and the most well-faceted surface was obtained at 1423 K. The insufficient driving force and slow crystal growth kinetics at 1123 K resulted in the characteristic of a terraced surface. In contrast, at 1523 K, the crystal surface further evolved into regularly curved facets, similar to the previously reported structure.⁷ The X-ray diffraction (XRD) patterns in Figures S9 and S10 demonstrate that all samples are indexed to the perovskite structure of STO and reveal the presence of trace amounts of SrCO₃ in all as-prepared samples. However, in the XRD patterns (Figures S11 and S12) of the corresponding postphotocatalytic samples, no distinct diffraction peaks attributed to SrCO₃ were detected, while obvious diffraction peaks associated with STO were observed. This indicates that SrCO₃ impurities are not present in the crystal bulk and would undergo decomposition during the photocatalytic reaction, which would not influence the photocatalytic activity (refer to the analysis in Figure S12 for details). Furthermore, a slight lattice shrinkage caused by Al doping is detected and the crystallinity of crystals improves with increasing temperature.³²

In wet chemical synthesis, surface capping agents including both organic and inorganic additives are used to selectively

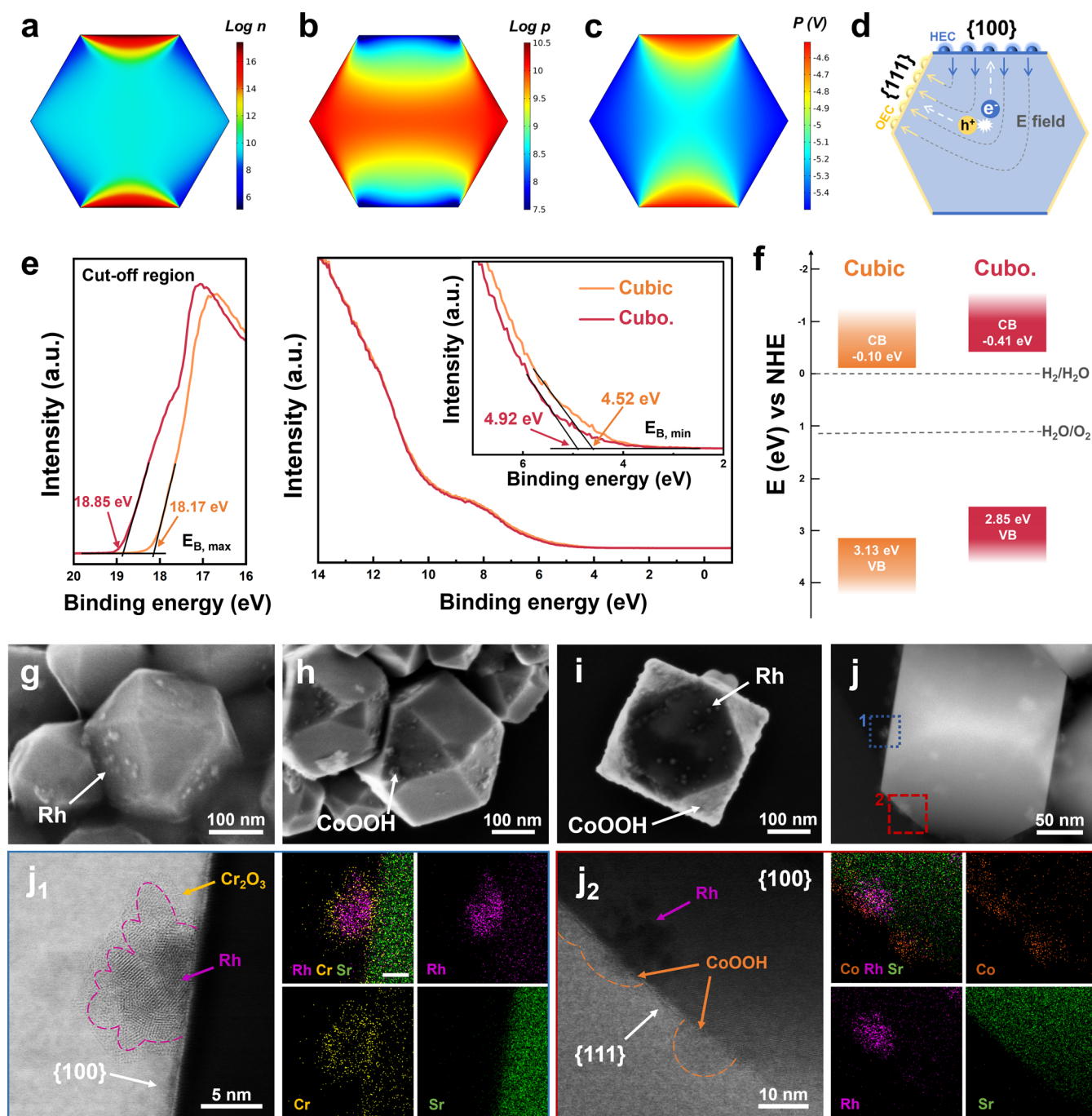


Figure 4. Photocarrier anisotropic separation among {100} and {111} facets. Simulated photocarrier distribution of (a) density of electrons (e^-), n , and (b) density of holes (h^+), p , at the cross section of Cubo-STO single crystals and (c) corresponding simulated potential distribution. (d) A schematic illustration of photocarrier separation driven by built-in electric fields at the cross section of a Cubo-STO single crystal loaded with hydrogen evolution cocatalysts (HEC) and oxygen evolution cocatalysts (OEC) under light illumination. (e) UPS data for the typical cubic and cuboctahedron (Cubo.) STO. Helium 1α ($h\nu = 21.22$ eV) spectra of secondary electron cutoff (left panel) and UPS spectra in the valence band (VB) region (right panel) are shown. The inset shows the enlarged UPS spectra in the VB region, and the intercept was noted as $E_{B,min}$. The intercept in the secondary electron cutoff (left panel) shows $E_{B,max}$ and the VBM was obtained by $h\nu - (E_{B,max} - E_{B,min})$. The position of CBM with respect to VBM was defined by the optical band gap (3.23 and 3.26 eV). (f) Energy-level scheme for the typical cubic and cuboctahedron STO based on the parameters derived from UPS spectra, the normal hydrogen electrode (NHE) value was derived by subtracting 4.44 ± 0.02 V from the vacuum level. (g–i) SEM images of Cubo-STO deposited with (g) 0.4 wt % Rh, (h) 0.4 wt % CoOOH, (i) 0.4 wt % Rh, and 0.4 wt % CoOOH. (j) HAADF-STEM image of Cubo-STO deposited with Rh (0.4 wt %)/Cr (0.2 wt %)/CoOOH (0.4 wt %) (j_1 , j_2) bright-field STEM images and corresponding EDS maps of blue box and red box in (j), suggesting a clear core/shell structure Rh/Cr₂O₃ nanoparticle loaded on the {100} facet and CoOOH loaded on the {111} facet.

tailor crystalline surfaces to obtain specific exposed facets.^{14,15,30,31} Herein, it is confirmed that in the salt melt environment, the precisely added inorganic ions (Al^{3+} , here)

doped at the surface region to kinetically control the thermally induced surface orientation growth of single crystals, exposing {111} facets and tailoring the crystal geometry structure.

However, distinct from solution system, the management of nanoscale processes of crystal morphology and structure transformation through solid-state synthesis is extremely difficult due to the intrinsically slow substance diffusion.^{33,34} A unique presynthesized precursor of low-crystalline STO nanospheres (Figure 3f) was adopted to decrease the atomic diffusion distances and kinetics barriers.^{27,34} Because the nanospheres are encapsulated by the molten salt medium, the crystal morphology evolution mainly occurs through a single-particle recrystallization process. For comparison, a series of nanosphere precursors (Figure S13) ranging from initial partial crystallization to well-crystallized mesoporous structures were prepared through a reverse crystal growth process.²⁷ Figures S14 and S15 illustrate the SEM images of corresponding samples after 10 h of calcination at 1173 and 1423 K, respectively. The cuboctahedron morphology can form at 1173 K; however, due to the insufficient driving force, the crystals can only evolve into mesoporous surfaces. As the temperature was elevated to 1423 K, the recrystallization process was facilitated such that the surface holes were eliminated, but the crystal surfaces were still rough. This suggests the strong chemical bonding of well-crystallized precursors restricts the flexible atomic migration, even under a strong thermal driving force at 1423 K. Thus, the low-crystalline nanospheres with a reaction time of 1–3 h efficiently promote solid-state transformation reaction kinetics to successfully synthesize well-defined single crystals with smooth surfaces. In addition, a cubic mesoporous STO (CM-STO) precursor (Figure S16a,b) with a reaction time of 48 h was synthesized through a hydrothermal reaction similar to that of Low-Cry-STO. XRD patterns (Figure S17) reveal that a small amount of SrCO_3 is also present in the CM-STO samples similar to the Low-Cry-STO precursors. Using this sample as a precursor, a series of single crystals with varying amounts of Al doping were prepared employing the same synthesis method as that for Cubo-STO single crystals. The as-prepared samples are indexed to the perovskite structure of STO. As shown in Figure S16c–h, within the range of Al^{3+} additive from 0.0 to 0.4 mol %, the STO single crystals synthesized using this precursor exhibit the same cubic morphology irrespective of Al doping, which is associated with its good crystallinity and inherent cubic shape. These results confirm the important role of Low-Cry-STO precursor in forming well-defined Cubo-STO.

The structural feature of the typical Low-Cry-STO precursor with a reaction time of 2 h was investigated in detail. The TEM images in Figure 3g–i show abundant irregular tiny microcrystals randomly dispersed in the disordered medium, suggesting that the particles were partially crystallized with both crystalline and disordered phases. The tiny microcrystal phase might be composed of medium-range ordered building blocks that are periodically stacked to tens of nanometers.³⁵ The XRD pattern of Low-Cry-STO in Figure S10 demonstrates the presence of perovskite STO crystalline phase, but the diffraction peak position is located at a much lower angle due to the inadequate crystallinity with the larger lattice parameters. Besides, the much lower intensity of the diffraction peaks indicates the massive presence of the disordered phase. It is believed that the Low-Cry-STO with a relatively weak bonding force kinetically boosts the atomic diffusion and mobility during recrystallization. Raman spectroscopy and X-ray absorption near-edge structure (XANES) spectroscopy (Figures S18 and S19) further confirm the distorted local

structure and weak chemical bonding of Low-Cry-STO precursors (refer to the analysis in Figures S18 and S19 for details). Therefore, the Low-Cry-STO particles are composed of microcrystals with distorted local structures dispersed in a major disordered medium.^{36–40} This low-crystallinity state effectively promotes the reaction kinetics during recrystallization, facilitating the transformation into single-crystal Cubo-STO.

All of the results above confirm an effective strategy for crystal facet engineering through the solid-state recrystallization reaction. A low-crystalline nanosphere precursor is synthesized to improve solid-state transformation reaction kinetics by shortening the atomic migration distance and weakening atomic diffusion resistance. The additive of Al^{3+} ions is adopted to tailor crystal surface structure to expose {111} facets with a controllable percentage. Consequently, driven by the high temperature and the strong polarization provided by the molten salt, low-crystalline nanoparticles can transform into highly crystallized single crystals with a well-faceted structure via solid-state recrystallization.

Anisotropic Photocarrier Separation between {100} and {111} Facets. The photocarrier distribution in a Cubo-STO particle was simulated using a simplified two-dimensional (2D) mode (Figure 4a–d). The anisotropic separations of electrons and holes are illustrated in Figure 4a,b, with an internal potential difference of 1 V between {100} and {111} facets (Figure 4c). The ultraviolet photoelectron spectroscopy (UPS) was conducted to determine the valence band maximum (VBM). As illustrated in Figure 4e, the VBM values were determined to be 3.13 and 2.85 eV versus the normal hydrogen electrode (NHE) for cubic and cuboctahedron STO, respectively. The conduction band minimum (CBM) was estimated by the band gap obtained from ultraviolet–visible diffuse reflectance spectroscopy (UV–vis DRS) in Figure S20, and the schematic energy levels were demonstrated in Figure 4f. The upward shift of VBM and CBM for Cubo-STO should be attributed to the change of the surface termination caused by {111} facets, and a p–n junction with a built-in electric field would be constructed between {100} and {111} facets.^{17,18} This indicates that photogenerated electrons and holes are thermodynamically favorable to transport to {100} and {111} facets driven by the built-in electric field (Figure 4d).

These results were further confirmed by in situ photo-deposition experiments. As shown in Figures 4g–i and S21, the photooxidation of $\text{CoOOH}/\text{MnO}_x$ deposition selectively occurred on the {111} facets, while the photoreduction of Rh was selectively deposited on the {100} facets. In pure water, the same deposition results on the {100} and {111} facets were displayed by adding Rh^{3+} and $\text{Co}^{2+}/\text{Mn}^{2+}$ precursors simultaneously. These results undoubtedly demonstrate the anisotropic separation of photoexcited carriers caused by the anisotropic facets. Furthermore, a 26-faceted STO single crystal enclosed with {100}, {110}, and {111} facets was synthesized for the determination of the redox reactivity ability. SEM images in Figure S21 show that when the three types of crystal facets coexisted in a single-crystal particle, the photooxidation of MnO_x is more likely to occur on the {111} facets. This phenomenon confirms our DFT results, showing that {111} facets have a more favorable hole-trapping ability. Based on the facts above, the core–shell-structured $\text{Rh}/\text{Cr}_2\text{O}_3$ and CoOOH cocatalysts were loaded as HEC and OEC by two-step photodeposition method with the sequence of simultaneous Rh and Co loading followed by Cr. Owing to

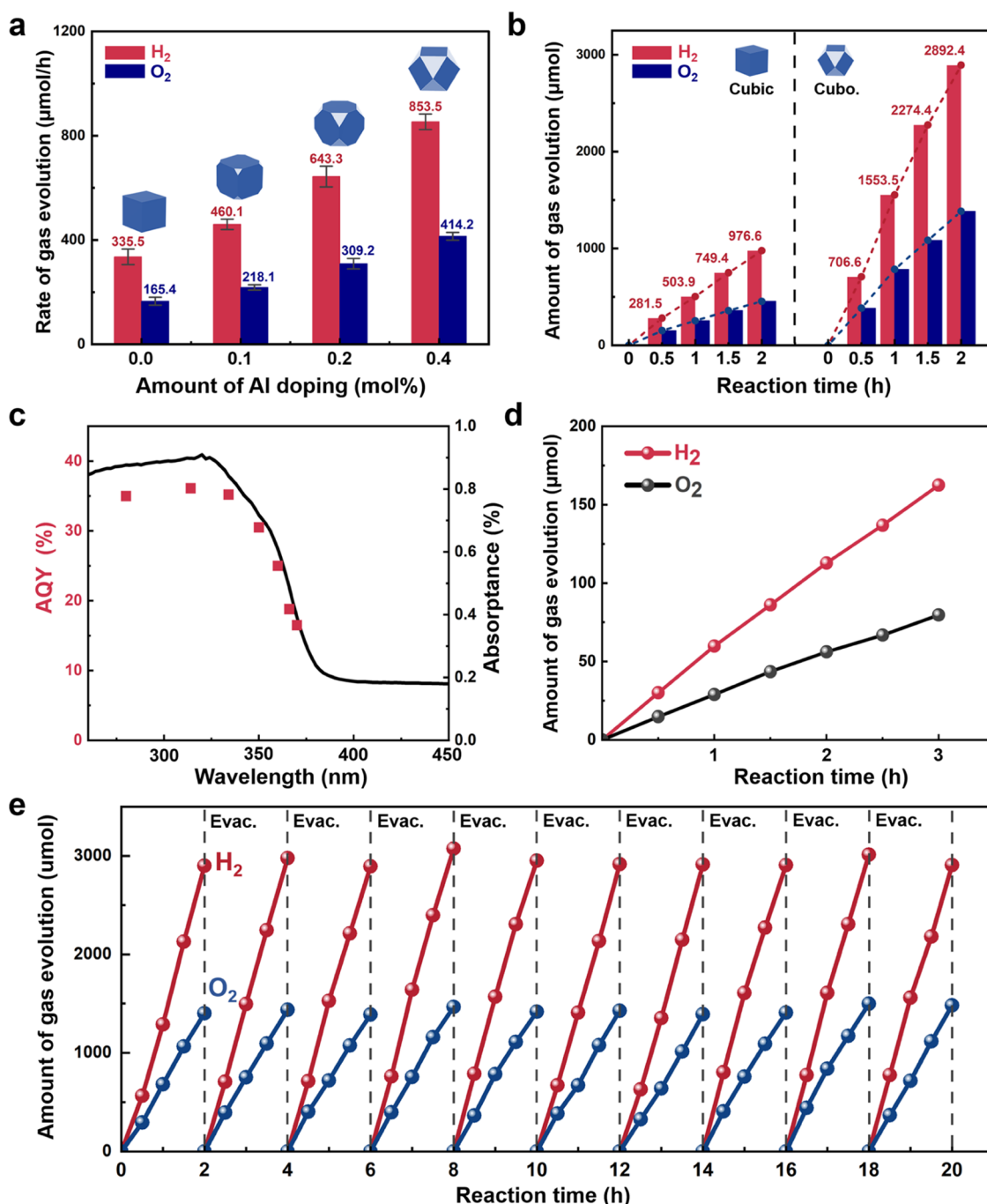


Figure 5. Overall water splitting performance of Rh/Cr₂O₃/CoOOH-modified cuboctahedron SrTiO₃. (a) Dependence of the water splitting activity on the amount of Al doping, indicating facet-dependent reactivity. Rh (0.1 wt %)/Cr₂O₃ (0.05 wt %)/CoOOH (0.1 wt %) was loaded and the amount of gas evolution at the point of the first hour during photocatalytic reaction for each sample was taken. (b) Time courses of gas evolution during overall water splitting over cubic and cuboctahedron STO modified with Rh (0.2 wt %)/Cr₂O₃ (0.1 wt %)/CoOOH (0.2 wt %). (c) The apparent quantum yield (AQY) of Cubo-STO plotted as a function of the wavelength of the incident light. (d) Reaction time course of Cubo-STO under simulated sunlight (AM 1.5G). (e) Stability test of Cubo-STO loaded with Rh (0.2 wt %)/Cr₂O₃ (0.1 wt %)/CoOOH (0.2 wt %) during photocatalytic water splitting. The produced H₂ and O₂ accumulated in the reaction system were evacuated every 2 h. The inner wall of the reactor window was cleaned after every evacuation. Reaction conditions: catalyst, 100 mg; reaction solution, 100 mL H₂O; light source, a 300 W Xe lamp (>300 nm) was employed in (a–c) and (e), while a solar simulator (AM 1.5G) was used in (b). For the measurement of AQY, the 300 W Xe lamp was equipped with various band-pass filters.

the well-defined crystal structure, HAADF-STEM imaging was employed to study the microstructure and distribution of the cocatalysts. The HAADF-STEM images and the corresponding EDS elemental maps in Figures 4j and S22 clearly show the core-shell structure of the loaded Rh/Cr₂O₃ nanoparticles on the {100} facets and the CoOOH species on the {111} facets.

A common photodeposition procedure⁷ in a sequence of Rh, Cr, and Co was also conducted. Figure S23 depicts a consistent photocatalytic OWS gas evolution rate, indicating the negligible influence of cocatalyst loading sequence on performance. This is attributed to the well-defined anisotropic

structure that determines the distribution for cocatalyst loading.

Photocatalytic Overall Water Splitting Performance Evaluation. The photocatalytic activity for the OWS was examined with 100 mg of photocatalysts dispersed in 100 mL of pure water after the deposition of the core-shell structure Rh/Cr₂O₃ HEC and CoOOH OEC under illumination by a 300 W Xe lamp. As shown in Figure S24a, the OWS reactivity was shown as a volcano curve with the elevated calcined temperature, and the maximum activity was reached at 1423 K. The electron paramagnetic resonance (EPR) spectra in Figure S25 indicate the temperature-dependent oxygen vacancies (V_O) concentration, which is related to the release of lattice oxygen.⁴¹ Moreover, UV-vis DRS spectroscopy in Figure S26 demonstrates that all of the samples exhibited an essentially constant absorption edge at 384 nm, but the crystals obtained at 1423 K show a much lower background of the light absorption beyond 400 nm, demonstrating its relatively low defect density. The higher background absorption at 1523 and 1223 K should be caused by abundant V_O and relatively poor crystallinity, respectively, which is unfavorable for activity.⁸ Thus, the Cubo-STO obtained at 1423 K shows the importance of high crystallization with low defect densities for OWS.

The photocatalytic OWS performance was found to follow a positive correlation with the exposure percentage of {111} facets (Figure 5a). The highest photocatalytic activity was achieved by Cubo-STO with the maximum fraction of {111} facet up to 36.6%, and the corresponding gas evolution rates related to reaction time are presented in Figure S24b. However, the effect of crystal facet exposure on the photocatalytic activity should be distinguished from the Al doping effect. Al doping has been reported to change the electronic structure of the photocatalysts, thereby influencing the OWS performance.⁴² X-ray photoelectron spectroscopy (XPS) spectra in Figure S27 reveal a coherent shift toward low binding energies for all of the peaks in Ti, O, and Sr core level spectra of Cubo-STO with respect to Low-Cry-STO. This general shift can be attributed to the shift in the Fermi level position, as brought about by a change in the depletion layer.^{42,43} Whereas no displacement is observed for all samples with 0.1–0.4 mol % Al³⁺ additive in Figure S28, suggesting that the addition of Al³⁺ below 0.4 mol % has little effect on the bulk electronic structure. Additionally, only when the addition is above 0.4 mol %, a concomitant lattice contraction can be observed from XRD patterns (Figure S9), which is caused by the substitution of smaller radius Al³⁺ for bulk Ti⁴⁺ sites.⁴³ Thus, the additives of Al³⁺ ions below 0.4 mol % are believed to mainly work on the surface to tailor the facet structure. When the Al³⁺ addition exceeds 0.4 mol %, excessive Al³⁺ dopes into the bulk, inducing lattice contraction. Thus, within the doping range of 0–0.4 mol %, the enhanced OWS performance should be ascribed to the better photocarrier separation resulting from the enlarged {111} facet but not bulk Al doping.

To better clarify the effect of crystal facet exposure, separating from Al doping, the photocatalytic OWS activity of cubic STO single crystals obtained from the CM-STO precursor was examined, which were loaded with an optimized mass of cocatalyst. Since the morphologies of the samples with Al doping amounts of 0.0–0.4 mol % are all cubic, we are able to independently evaluate the substantive effect of Al doping on the OWS activity, excluding the influence of crystal facet

exposure. As shown in Figure S29, the reactivity is much lower compared with that of Cubo-STO and the performance improvement resulting from varying the Al additive within the range of 0.0–0.4 mol % is extremely limited. Therefore, we can conclude that the mechanism of performance enhancement is attributed to the efficient anisotropic separation of photo-excited carriers by {111} facet exposure. Time-resolved photoluminescence (TR-PL) emission decay spectroscopy in Figure S30 reveals an almost unchanged average lifetime from 9.53 to 9.62 ns. Despite the exposed {111} facet of Cubo-STO does not significantly extend carrier lifetimes compared with cubic STO, the photocatalytic performance of OWS has still been substantially enhanced, indicating that the reactivity does not depend on prolonged carrier lifetimes.⁴⁴ Instead, the highly efficient anisotropic charge separation and extraction mechanism between {100} and {111} facets plays a dominant role in the enhancement of photocatalytic OWS performance.

As shown in Figure S31, the photocatalytic activity for OWS can be further improved by increasing cocatalyst loading mass to Rh (0.2 wt %)/Cr₂O₃ (0.1 wt %)/CoOOH (0.2 wt %), which results from the enhancement of built-in electric field.¹⁸ Owing to the further enhancement of the anisotropic charge separation ability by cocatalysts, the H₂ evolution rate of 1553.5 $\mu\text{mol}\cdot\text{h}^{-1}$ was achieved, which is 3 times better than the cubic STO (Figure 5b). The ratios of H₂ and O₂ are close to the stoichiometric ratio expected from the splitting of H₂O into H₂ and O₂. Figure 5c presents the apparent quantum yield (AQY) values for the photocatalyst during overall water splitting as a function of the irradiation wavelength, which were determined to be 35.2% at 335 nm (± 10 nm), 30.5% at 350 nm (± 10 nm), and 24.9% at 360 nm (± 10 nm), respectively. Figure 5d shows a reaction time course of OWS under illumination of AM 1.5G simulated sunlight, and the corresponding solar to hydrogen (STH) efficiency was calculated to be 0.14%. The photocatalytic OWS activity of Cubo-STO remains stable after 10 cycles within 20 h (Figure 5e), demonstrating its superior stability.

CONCLUSIONS

In this work, we have successfully fabricated uniform SrTiO₃ single crystals exposing tailored {111} facets via a solid-state recrystallization reaction. A novel crystal facet engineering strategy through solid-state synthesis was developed: a low-crystalline SrTiO₃ precursor was used to improve the solid-state transformation reaction kinetics, and the additive of inorganic Al³⁺ ions was adopted to engineer the crystal surface orientation. Accordingly, the cuboctahedral SrTiO₃ single crystals enable the effective anisotropic separation of photo-excited electron–hole pairs between {100} and {111} facets. The photocatalytic OWS performance increases with the exposure percentage of {111} facets. Owing to the high crystallinity, low defect density, and anisotropic surface structures, the hydrogen evolution rate for the OWS of Cubo-STO reaches 1.55 mmol·h^{−1}, 3 times higher than isotropic SrTiO₃ enclosed with {100} facets. The proposed strategy will open new avenues to fabricate well-defined, high-crystalline single crystals enabling highly efficient solar fuel, solar power, (photo)catalysis, and energy storage technologies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c12062>.

Full detailed experimental section, theoretical calculation details, additional characterizations, photocatalytic measurements, and supporting results (PDF)

Su-Huai Wei — Beijing Computational Science Research Center, Beijing 100193, China; orcid.org/0000-0003-1563-4738

Complete contact information is available at:
<https://pubs.acs.org/10.1021/jacs.3c12062>

AUTHOR INFORMATION

Corresponding Authors

Peng Fei Liu — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China; orcid.org/0000-0003-0411-0488; Email: pfliu@ecust.edu.cn

Sheng Dai — Key Laboratory for Advanced Materials and Feringa Nobel Prize Scientist Joint Research Center, Institute of Fine Chemicals School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China; Email: shengdai@ecust.edu.cn

Hua Gui Yang — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China; orcid.org/0000-0003-0436-8622; Email: hgyang@ecust.edu.cn

Authors

Yang Zhang — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China

Xuefeng Wu — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China

Zhi-Hao Wang — Beijing Computational Science Research Center, Beijing 100193, China; orcid.org/0000-0002-6376-6519

Yu Peng — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China

Yuanwei Liu — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China

Shuang Yang — Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China; orcid.org/0000-0002-8244-3002

Chenghua Sun — Department of Chemistry and Biotechnology, Faculty of Science, Engineering and Technology, Swinburne University of Technology, Hawthorn 3122, Australia; orcid.org/0000-0001-7654-669X

Xiaoxiang Xu — Shanghai Key Lab of Chemical Assessment and Sustainability, School of Chemical Science and Engineering, Tongji University, Shanghai 200092, China

Xie Zhang — Beijing Computational Science Research Center, Beijing 100193, China; School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an 710072, China; orcid.org/0000-0003-3424-9253

Jun Kang — Beijing Computational Science Research Center, Beijing 100193, China; orcid.org/0000-0003-4788-0028

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was financially supported by the National Key Research and Development Program of China (No: 2023YFA1507102), the National Natural Science Foundation of China (51920105003, 22239001, 12088101, 11991060, 52172136, 22376062, and U2230402), the Shanghai Pilot Program for Basic Research (22TQ1400100-12), the Science and Technology Commission of Shanghai Municipality (21DZ1207101 and 22ZR1415700), and the Innovation Program of Shanghai Municipal Education Commission (E00014). The authors also thank the Frontiers Science Center for Materiobiology and Dynamic Chemistry, the crew of the BL14W1 beamline at the Shanghai Synchrotron Radiation Facility (SSRF), and the 1W1B beamline of Beijing Synchrotron Radiation Facility (BSRF) for their constructive assistance with the XAFS measurements and data analyses.

REFERENCES

- (1) Kim, J. H.; Hansora, D.; Sharma, P.; Jang, J. W.; Lee, J. S. Toward Practical Solar Hydrogen Production - an Artificial Photosynthetic Leaf-to-farm Challenge. *Chem. Soc. Rev.* **2019**, *48*, 1908–1971.
- (2) Nishiyama, H.; Yamada, T.; Nakabayashi, M.; Maehara, Y.; Yamaguchi, M.; Kuromiya, Y.; Nagatsuma, Y.; Tokudome, H.; Akiyama, S.; Watanabe, T.; Narushima, R.; Okunaka, S.; Shibata, N.; Takata, T.; Hisatomi, T.; Domen, K. Photocatalytic Solar Hydrogen Production From Water on a 100m² Scale. *Nature* **2021**, *598*, 304–307.
- (3) Hisatomi, T.; Domen, K. Reaction Systems for Solar Hydrogen Production via Water Splitting With Particulate Semiconductor Photocatalysts. *Nat. Catal.* **2019**, *2*, 387–399.
- (4) Zhou, P.; Navid, I. A.; Ma, Y.; Xiao, Y.; Wang, P.; Ye, Z.; Zhou, B.; Sun, K.; Mi, Z. Solar-to-hydrogen Efficiency of More Than 9% in Photocatalytic Water Splitting. *Nature* **2023**, *613*, 66–70.
- (5) Fujishima, A.; Honda, K. Electrochemical Photolysis of Water at a Semiconductor Electrode. *Nature* **1972**, *238*, 37–38.
- (6) Wang, Q.; Domen, K. Particulate Photocatalysts for Light-driven Water Splitting: Mechanisms, Challenges, and Design Strategies. *Chem. Rev.* **2020**, *120*, 919–985.
- (7) Takata, T.; Jiang, J.; Sakata, Y.; Nakabayashi, M.; Shibata, N.; Nandal, V.; Seki, K.; Hisatomi, T.; Domen, K. Photocatalytic Water Splitting With a Quantum Efficiency of Almost Unity. *Nature* **2020**, *581*, 411–414.
- (8) Wang, Z.; Inoue, Y.; Hisatomi, T.; Ishikawa, R.; Wang, Q.; Takata, T.; Chen, S.; Shibata, N.; Ikuhara, Y.; Domen, K. Overall Water Splitting by Ta₃N₅ Nanorod Single Crystals Grown on The Edges of KTaO₃ Particles. *Nat. Catal.* **2018**, *1*, 756–763.
- (9) Wang, Q.; Nakabayashi, M.; Hisatomi, T.; Sun, S.; Akiyama, S.; Wang, Z.; Pan, Z.; Xiao, X.; Watanabe, T.; Yamada, T.; et al. Oxy sulfide Photocatalyst for Visible-light-driven Overall Water Splitting. *Nat. Mater.* **2019**, *18*, 827–832.
- (10) Wang, X.; Maeda, K.; Thomas, A.; Takanabe, K.; Xin, G.; Carlsson, J. M.; Domen, K.; Antonietti, M. A metal-free Polymeric Photocatalyst for Hydrogen Production from Water Under Visible Light. *Nat. Mater.* **2009**, *8*, 76–80.

- (11) Song, X.; Wei, G.; Sun, J.; Peng, C.; Yin, J.; Zhang, X.; Jiang, Y.; Fei, H. Overall Photocatalytic Water Splitting by an Organolead Iodide Crystalline Material. *Nat. Catal.* **2020**, *3*, 1027–1033.
- (12) Chen, R.; Ren, Z.; Liang, Y.; Zhang, G.; Ditttrich, T.; Liu, R.; Liu, Y.; Zhao, Y.; Pang, S.; An, H.; Ni, C.; Zhou, P.; Han, K.; Fan, F.; Li, C. Spatiotemporal Imaging of Charge Transfer in Photocatalyst Particles. *Nature* **2022**, *610*, 296–301.
- (13) Wang, D.; Sheng, T.; Chen, J.; Wang, H.-F.; Hu, P. Identifying The Key Obstacle in Photocatalytic Oxygen Evolution on Rutile TiO₂. *Nat. Catal.* **2018**, *1*, 291–299.
- (14) Wang, S.; Liu, G.; Wang, L. Crystal Facet Engineering of Photoelectrodes for Photoelectrochemical Water Splitting. *Chem. Rev.* **2019**, *119*, 5192–5247.
- (15) Yang, H. G.; Sun, C. H.; Qiao, S. Z.; Zou, J.; Liu, G.; Smith, S. C.; Cheng, H. M.; Lu, G. Q. Anatase TiO₂ Single Crystals with a Large Percentage of Reactive Facets. *Nature* **2008**, *453*, 638–641.
- (16) Yu, J.; Low, J.; Xiao, W.; Zhou, P.; Jaroniec, M. Enhanced Photocatalytic CO₂-reduction Activity of Anatase TiO₂ by Coexposed {001} and {101} Facets. *J. Am. Chem. Soc.* **2014**, *136*, 8839–8842.
- (17) Luo, Z.; Ye, X.; Zhang, S.; Xue, S.; Yang, C.; Hou, Y.; Xing, W.; Yu, R.; Sun, J.; Yu, Z.; Wang, X. Unveiling the Charge Transfer Dynamics Steered by Built-in Electric Fields in BiOBr Photocatalysts. *Nat. Commun.* **2022**, *13*, No. 2230.
- (18) Chen, R.; Fan, F.; Li, C. Unraveling Charge-separation Mechanisms in Photocatalyst Particles by Spatially Resolved Surface Photovoltage Techniques. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202117567.
- (19) Tachikawa, T.; Yamashita, S.; Majima, T. Evidence for Crystal-face-dependent TiO₂ Photocatalysis from Single-molecule Imaging and Kinetic Analysis. *J. Am. Chem. Soc.* **2011**, *133*, 7197–7204.
- (20) Mu, L.; Zhao, Y.; Li, A.; Wang, S.; Wang, Z.; Yang, J.; Wang, Y.; Liu, T.; Chen, R.; Zhu, J.; Fan, F.; Li, R.; Li, C. Enhancing Charge Separation on High Symmetry SrTiO₃ Exposed with Anisotropic Facets for Photocatalytic Water Splitting. *Energy Environ. Sci.* **2016**, *9*, 2463–2469.
- (21) Mao, X.; Chen, P. Inter-facet Junction Effects on Particulate Photoelectrodes. *Nat. Mater.* **2022**, *21*, 331–337.
- (22) Ma, C. Q.; Eickemeyer, F. T.; Lee, S. H.; Kang, D. H.; Kwon, S. J.; Gratzel, M.; Park, N. G. Unveiling Facet-dependent Degradation and Facet Engineering for Stable Perovskite Solar Cells. *Science* **2023**, *379*, 173–178.
- (23) Avcioglu, C.; Avcioglu, S.; Bekheet, M. F.; Gurlo, A. Photocatalytic Overall Water Splitting by SrTiO₃: Progress Report and Design Strategies. *ACS Appl. Energy Mater.* **2023**, *6*, 1134–1154.
- (24) Zhao, Z.; Li, Z.; Zou, Z. A. Theoretical Study of Water Adsorption and Decomposition on The Low-index Stoichiometric Anatase TiO₂ Surfaces. *J. Phys. Chem. C* **2012**, *116*, 7430–7441.
- (25) Giocondi, J. L.; Rohrer, G. S. Structure Sensitivity of Photochemical Oxidation and Reduction Reactions on SrTiO₃ Surfaces. *J. Am. Ceram. Soc.* **2003**, *86*, 1182–1189.
- (26) Jia, Q.; Wang, C.; Liu, J.; Cai, X.; Zhong, L.; Chen, S.; Li, T.; Yu, G.; Wu, L. Z.; Duan, D. Synergistic Effect of Sr-O Divacancy and Exposing Facets in SrTiO₃ Micro/nano Particle: Accelerating exciton Formation and Splitting, Highly Efficient Co²⁺ Photooxidation. *Small* **2022**, *18*, No. 2202659.
- (27) Zhu, J. P.; Wu, X. F.; Yang, X. H.; Liu, P. F.; Dai, S.; Yang, H. G. A Template-free Synthesis of Mesoporous SrTiO₃ Single Crystals. *CrystEngComm* **2021**, *23*, 5595–5600.
- (28) Shibata, N.; Findlay, S. D.; Kohno, Y.; Sawada, H.; Kondo, Y.; Ikuhara, Y. Differential Phase-contrast Microscopy at Atomic Resolution. *Nat. Phys.* **2012**, *8*, 611–615.
- (29) Kumar, A.; Dutta, S.; Kim, S.; Kwon, T.; Patil, S. S.; Kumari, N.; Jeevanandham, S.; Lee, I. S. Solid-state Reaction Synthesis of Nanoscale Materials: Strategies and Applications. *Chem. Rev.* **2022**, *122*, 12748–12863.
- (30) Yang, S.; Yang, B. X.; Wu, L.; Li, Y. H.; Liu, P.; Zhao, H.; Yu, Y. Y.; Gong, X. Q.; Yang, H. G. Titania Single Crystals with a Curved Surface. *Nat. Commun.* **2014**, *5*, No. 5355.
- (31) Liu, G.; Yang, H. G.; Pan, J.; Yang, Y. Q.; Lu, G. Q.; Cheng, H. M. Titanium Dioxide Crystals with Tailored Facets. *Chem. Rev.* **2014**, *114*, 9559–9612.
- (32) Qin, Y.; Fang, F.; Xie, Z.; Lin, H.; Zhang, K.; Yu, X.; Chang, K. La, Al-Codoped SrTiO₃ as a La, Al-codoped SrTiO₃ As a Photocatalyst in Overall Water Splitting: Significant Surface Engineering Effects on Defect Engineering. *ACS Catal.* **2021**, *11*, 11429–11439.
- (33) Chen, J.; Jiang, F.; Yin, Y. Manipulation of Interfacial Diffusion for Controlling Nanoscale Transformation. *Acc. Chem. Res.* **2021**, *54*, 1168–1177.
- (34) Li, L.; Deng, J.; Chen, J.; Xing, X. Topochemical Molten Salt Synthesis for Functional Perovskite Compounds. *Chem. Sci.* **2016**, *7*, 855–865.
- (35) Lan, S.; Zhu, L.; Wu, Z.; Gu, L.; Zhang, Q.; Kong, H.; Liu, J.; Song, R.; Liu, S.; Sha, G.; Wang, Y.; Liu, Q.; Liu, W.; Wang, P.; Liu, C.; Ren, Y.; Wang, X. A Medium-range Structure Motif Linking Amorphous and Crystalline States. *Nat. Mater.* **2021**, *20*, 1347–1352.
- (36) Vračar, M.; Kuzmin, A.; Merkle, R.; Purans, J.; Kotomin, E. A.; Maier, J.; Mathon, O. Jahn-Teller Distortion Around Fe⁴⁺ in Sr(Fe_xTi_{1-x})O_{3-δ} from X-ray Absorption Spectroscopy, X-ray Diffraction, and Vibrational Spectroscopy. *Phys. Rev. B* **2007**, *76*, No. 174107.
- (37) Tenne, D. A.; Gonenli, I. E.; Soukiasian, A.; Schlom, D. G.; Nakhmanson, S. M.; Rabe, K. M.; Xi, X. X. Raman Study of Oxygen Reduced and Re-oxidized Strontium Titanate. *Phys. Rev. B* **2007**, *76*, No. 024303.
- (38) Frenkel, A. I.; Ehre, D.; Lyahovitskaya, V.; Kanner, L.; Wachtel, E.; Lubomirsky, I. Origin of Polarity in Amorphous SrTiO₃. *Phys. Rev. Lett.* **2007**, *99*, No. 215502.
- (39) Ghaffari, M.; Liu, T.; Huang, H.; Tan, O. K.; Shannon, M. Investigation of Local Structure Effect and X-ray Absorption Characteristics (EXAFS) of Fe (Ti) K-edge on Photocatalyst Properties of SrTi_{1-x}Fe_xO_{3-δ}. *Mater. Chem. Phys.* **2012**, *136*, 347–357.
- (40) Kodre, A.; Arčon, I.; Gomilšek, J. P. n.; Zalar, B. EXAFS Study of Disorder in SrTiO₃ Perovskite. *AIP Conf. Proc.* **2007**, *882*, 481–483.
- (41) Takata, T.; Domen, K. Defect Engineering of Photocatalysts by Doping of Aliovalent Metal Cations for Efficient Water Splitting. *J. Phys. Chem. C* **2009**, *113*, 19386–19388.
- (42) Zhao, Z.; Goncalves, R. V.; Barman, S. K.; Willard, E. J.; Byle, E.; Perry, R.; Wu, Z.; Huda, M. N.; Moulé, A. J.; Osterloh, F. E. Electronic Structure Basis for Enhanced Overall Water Splitting Photocatalysis with Aluminum Doped SrTiO₃ in Natural Sunlight. *Energy Environ. Sci.* **2019**, *12*, 1385–1395.
- (43) Moss, B.; Wang, Q.; Butler, K. T.; Grau-Crespo, R.; Selim, S.; Regoutz, A.; Hisatomi, T.; Godin, R.; Payne, D. J.; Kafizas, A.; Domen, K.; Steier, L.; Durrant, J. R. Linking In Situ Charge Accumulation to Electronic Structure in Doped SrTiO₃ Reveals Design Principles for Hydrogen-evolving Photocatalysts. *Nat. Mater.* **2021**, *20*, 511–517.
- (44) Li, R.; Takata, T.; Zhang, B.; Feng, C.; Wu, Q.; Cui, C.; Zgabf, Z.; Domen, K.; Li, Y. Criteria for Efficient Photocatalytic Water Splitting Revealed by Studying Carrier Dynamics in a Model Al-Doped SrTiO₃ Photocatalyst. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202313537.